

CVE-2026-31431

Copy Fail at the Edge

NVIDIA Jetson runs Ubuntu. Ubuntu is vulnerable.
Every edge AI device in the field is exposed.

The case for Red Hat Device Edge

THE VULNERABILITY

< 1 sec

Unprivileged user to root. Deterministic. 732 bytes of Python.

Zero forensic trace.

No disk modification. No audit log. No file integrity alert. Evidence destroyed on reboot. In the kernel since 2017.

CVSS 7.8 HIGH • Red Hat Bugzilla 2460538

"Local only" is not a shield at the edge.

Compromised container on the device → **local access**

Vulnerable inference API or exposed SSH → **local access**

Physical access to field hardware → **local access**

Supply-chain compromise of a model or runtime → **local access**

Any of these converts to root in under one second. No evidence left behind.

DIRECT EXPOSURE

NVIDIA Jetson is vulnerable.

Jetson runs L4T (Linux for Tegra) – a full Ubuntu kernel. The vulnerable crypto module is enabled in the default config.

JETPACK	KERNEL	BASE OS	STATUS
JetPack 6.x (Orin)	5.15 / 6.1	Ubuntu 22.04	Vulnerable
JetPack 5.x (Xavier / Orin)	5.10	Ubuntu 20.04	Vulnerable
JetPack 4.x (Nano / TX2)	4.9	Ubuntu 18.04	Below range

You cannot patch what you do not control.

NVIDIA controls the L4T kernel. Customers cannot independently patch it.

Every day between disclosure and the next JetPack BSP release is a day every Jetson device in the field is exploitable with a publicly available script.

Data center: vendor errata, `dnf update`, hours.

Edge (Jetson): wait for NVIDIA BSP → requalify → field rollout → weeks to months.

WHAT THESE DEVICES DO

Production systems on OT networks.

SECTOR	EDGE AI WORKLOADS	ROOT =
Transportation	Wayside monitoring, defect detection, safety systems	Safety manipulation, regulatory exposure
Utilities	Substation monitoring, grid analytics	SCADA lateral movement, NERC CIP violations
Manufacturing	Quality inspection, robotic guidance	Production sabotage, IP exfiltration
Healthcare	Medical imaging, patient monitoring	HIPAA breach, patient safety risk

The lateral movement path every framework warns about.

Edge AI devices pull model updates over IP (**IT side**) and run inference that feeds operational decisions (**OT side**).

Copy Fail turns any IT-side foothold into OT-side root access — with no forensic trail.

**IT foothold → Copy Fail → OT root.
No trace. No alert. No evidence.**

Containers do not protect you here.

The Linux page cache is a host-level resource shared across all namespaces. Copy Fail corrupts a setuid binary in the page cache from inside a container.

NVIDIA recommends containerized inference on Jetson.

DeepStream, Triton, custom models – all in containers with Docker defaults.

Docker on Jetson: no seccomp filter for `AF_ALG`. No SELinux. No Falco.

A compromised model-serving container owns the host.

Defense in depth on the same hardware.

UBUNTU L4T

Kernel gated by NVIDIA BSP

RED HAT DEVICE EDGE

Red Hat errata on Red Hat's cadence

UBUNTU L4T

`apt upgrade` – mutable, no rollback

RED HAT DEVICE EDGE

`bootc` image – atomic, rollback-capable

UBUNTU L4T

Docker default – no seccomp for AF_ALG

RED HAT DEVICE EDGE

Podman + custom seccomp – AF_ALG blocked

SELinux vs. AppArmor at fleet scale.

APPARMOR (UBUNTU)

Per-profile

`deny network alg,` – must update every application profile individually

SELINUX (RHEL)

One custom policy module

denies `alg_socket`
`{ create }` for all confined domains – system-wide, one command

Plus: RHACS runtime detection at the edge. Red Hat Satellite + Ansible Automation Platform for fleet patching. FIPS 140-3 validated. STIG content. Red Hat Lightspeed CVE advisory.

None of these exist in the Ubuntu L4T stack.

Five steps.

- 1 Inventory all Jetson deployments.** JetPack 5.x or 6.x = vulnerable.
- 2 Blacklist the module.** If `algif_aead` is a module (not built-in), blacklist it today.
- 3 Assess containerized inference.** Docker defaults provide zero protection.
- 4 Evaluate Red Hat Device Edge.** SELinux + bootc + Podman + independent patching.
- 5 Plan for the next one.** This was hiding for nine years. Your edge devices need to respond when the next one is found.

You cannot patch what you do not control.

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Red Hat Enterprise Linux • Red Hat Device Edge • Red Hat OpenShift
Red Hat Advanced Cluster Security • Red Hat Satellite • Ansible Automation
Platform